cadence

EMAC Firmware Driver Quick Start Guide

Product Version 1.6.0

September 2017

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1. Overview

This document describes the basics of how to get up and running with the Cadence IP Core Driver.

- Prepare the Cadence Platform Services (CPS)
- Prepare interrupt handling
- Build the core-driver
- Use the core-driver to initialise the Cadence IP
- Implement additional features using the core-driver

2. Details

Here's how to get started:

1. *Prepare the Cadence Platform Services:*

   This is covered in detail in the Cadence Driver Porting Guide, which is available at doc/porting/porting_guide.pdf. A sample implementation for bare-metal systems is available in reference_code/tests/cps_emac_bm, and one for linux is supplied at reference_code/lnx_mod/src/cps_v2_linux.c. You may be able to use these without modification.

   CPS is a platform-specific code to connect a given Cadence driver to the rest of your system. Not all of the features of CPS are required for this driver. You only need to implement the functions for uncached access to memory:

   ```
   CPS_UncachedRead16, CPS_UncachedRead32,
   CPS_UncachedWrite16 and CPS_UncachedWrite32
   ```

   These functions will be used by the core driver to access the memory in a portable manner. Note that the core driver only uses 32-bit accesses, but the sample tests also use 16-bit accesses. CPS_CacheFlush() / CPS_CacheInvalidate() are not used by this driver.

   After this step you should have an object file implementing the CPS for your target system which you can link with later.

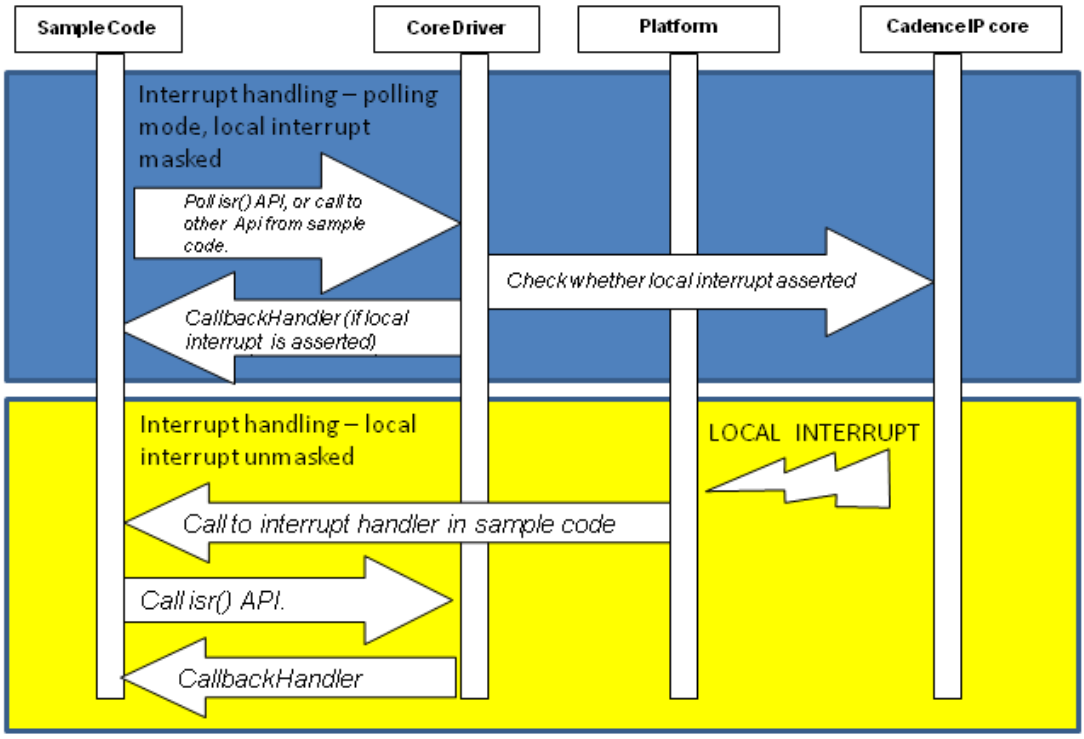
2. *Prepare interrupt handling*

   The EMAC interrupts can be handled either by polling or by interrupt handling, depending on the requirements of the system. If the EMAC 'ethernet_int' interrupt signal is not connected to

an interrupt controller, or remains disabled in the controller, the status of the IP core interrupts is polled as shown in the diagram below. The driver's *isr* API should be polled regularly to determine if there are any local interrupts active, and this will call the relevant callback functions to handle any active interrupts if they have been enabled. You can register callback functions when initialising the driver, these will be enabled when the *start* API is called.

If the EMAC interrupt is to be handled by a processor interrupt, then an interrupt handler should be written and hooked up to the 'ethernet_int' output. This handler should then call the driver's interrupt handler *isr*, which will in turn call any registered callback functions. Note that currently the *isr* API function handles interrupts from all priority queues, so the separate 'ethernet_int_q*' outputs should be OR'd together with 'ethernet_int', to generate a single processor interrupt for multi-queue operation. Details of how to hook up the interrupt handler will be processor and operating system specific, and are not covered in this documentation.

Figure 1. Local Interrupts



3. *Build the core-driver*

   The driver and a sample makefile is provided in the core_driver directory. You will need to customise the makefile for your build environment, e.g. compiler, any necessary libraries etc.

   After this step you should have an object file edd.o which can be linked with your own code.

4. *Use the core-driver to initialise the Cadence IP*

Sample code is available in the reference_code/tests directory. The sample code is designed to require only minor changes before it can be used. You will however need to change some default values in edd_test_supp.h to match your hardware, i.e.:

```
/* IP Base addresses - re-define as appropriate
   for your hardware platform */
#define EMAC_REG_BASE_ADDRESS0   (0xFFF30000)
#define EMAC_REG_BASE_ADDRESS1   (0xFFF32000)
```

The sample code includes several test functions in mac_test.c, as listed in the README.txt file, which are called from the main() function in example_main.c. Most of the tests rely on a test environment which has two EMAC IP instances connected to each other, which is why two base addresses are defined. However, loopbackTest requires only a single EMAC instance, using the EMAC_REG_BASE_ADDRESS0 address definition.

Each test calls a test setup function which creates and initialises a driver instance. This first uses *probe* to determine the size of memory which is required for private data. The memory must be provided by the code calling the driver, and may be allocated at compile time (e.g. by an array allocation) or dynamically. This private data will be passed to the driver as a parameter for each call to the API.

The sample code then prepares some example initialisation parameters, in a CEDI_Config struct, e.g. Tx and Rx rings and IP configuration settings, etc. You should replace these with any default settings you wish to configure: please see the user guide at doc/emac_driver/ emac_driver_guide.pdf for more details on all available options.

The sample code then calls *init* to initialise the driver and the hardware. The test function continues by preparing Rx buffers, calling *start* to enable the IP interrupts (or "events"), and enabling the Tx and Rx EMAC functionality.

During the tests the sample code displays some diagnostic information using stdio via the 'cInst->printf' call which is defined in example_main.c. If stdio is not available on your system, the env_printf operation should be defined as NULL or an alternative (e.g. write to memory or local storage.)

After modifying the sample code to suit your environment, you should compile for your target system and link with the driver and CPS objects prepared previously, then download and run the code on your target system. This will be hardware and operating system specific, and is not covered in this documentation. An example makefile suitable for building the driver and reference test code on a linux system is provided.